

AI-Powered New-Hire Onboarding Concierge

Project Prompt Document

Tools Used: GitHub Copilot & Visual Studio Code

Objective

When a new employee joins, the system should automatically start an onboarding workflow and track onboarding progress across 8 tasks. The project should demonstrate an orchestrator agent that coordinates multiple tools and updates task statuses for managers.

Architecture Decisions

Orchestration: Pure Rule-Based Orchestration (No LLM required for workflow execution).

Orchestrator Implementation: Python business logic and workflow management.

Database: SQLite as the primary database.

Authentication: JWT-based custom authentication.

Seeded Accounts:

HR: hr@company.com / Hr@123

Manager: manager@company.com / Manager@123

UI Theme: Light Corporate Enterprise

Primary color: Blue

Completed tasks: Green

In-progress tasks: Orange

Failed tasks: Red

Responsive modern enterprise dashboard styling

Core Workflow

1. HR logs into the system.
2. HR creates a new employee record by entering: Employee Name, Employee ID, Email, Department, Designation, and Joining Date.
3. When HR clicks "Start Onboarding", a simulated "New Joiner" webhook event is triggered.
4. The Orchestrator Agent receives the event and executes the following 8 onboarding tasks sequentially:
5. Each onboarding task must be implemented as an independent Python function (mock tool).
6. The orchestrator agent must call each tool in order, track task status, record timestamps, update onboarding progress percentage, log every action, handle failures gracefully, and continue or stop execution based on workflow rules.

The 8 Sequential Onboarding Tasks:

1. Create Employee Account

2. Create Company Email
3. Request Laptop
4. Assign Software Licenses
5. Generate Employee ID Card
6. Add Employee to Team Workspace
7. Grant System Access Permissions
8. Register Employee in HR Database

Task Execution Behavior

Simulate realistic onboarding processing.

Each task should take approximately 1 to 2 seconds.

Show real-time progress updates.

Introduce occasional failures with a low probability (around 10%).

Failed tasks should display meaningful failure reasons.

Allow managers to retry failed tasks.

Display execution history and logs.

User Roles

HR Role

Create employee records.

Trigger onboarding workflow.

View onboarding status.

Manager Role

View onboarding dashboard.

Monitor onboarding progress.

Track task execution.

Retry failed tasks.

View execution logs and history.

Pages Required

1. Login Page

Email

Password

Role Selection

JWT Authentication

2. Dashboard

Total Employees

Active Onboarding Processes

Completed Onboardings
Failed Tasks
Recent Activity

3. New Employee Page

Employee onboarding form
Start Onboarding button

4. Employee Status Page

Employee details
Progress bar
Task timeline
Completion percentage
Execution logs

5. Manager Monitoring Page

Employee list
Search and filter
Real-time onboarding status
Failed task management
Retry functionality

Database Requirements

SQLite is used as the primary database with the following tables:

Employees Table

employee_id
name
email
department
designation
joining_date
onboarding_status

Tasks Table

task_id
employee_id
task_name
status
started_at
completed_at
error_message

Logs Table

log_id
employee_id
action
timestamp

Technical Requirements

Language: Python
Framework: FastAPI (Flask acceptable)
Database: SQLite
Authentication: JWT
Modular architecture
Responsive UI with professional enterprise design
Status badges: Pending, Running, Completed, Failed
Progress indicators
Real-time updates
Simulated webhook trigger

Orchestrator Agent Requirements

Act as a workflow coordinator.
Execute onboarding tasks sequentially.
Update database after each task.
Maintain execution logs.
Store workflow history.
Expose execution data to managers.
Demonstrate agent orchestration clearly.

Success Criteria

A manager should be able to create a new employee onboarding request, trigger the onboarding workflow, observe all 8 onboarding tasks executing in real time, view progress updates, monitor logs, handle failures, retry tasks, and track onboarding completion through a professional enterprise dashboard.